What is needed is an assumption that the level sets $\hat{\Sigma}_\alpha = \{\boldsymbol{q} | U(\boldsymbol{q}) = \alpha\}$ are bounded uniformly for $\alpha \in [U_{\min}, \mathrm{E}_0]$. Then it follows that solutions satisfying the energy constraint remain confined to a compact (closed and bounded) set.

The assumption on U is not satisfied by some simple potentials. For example consider $U(x, y) = x^2$ which is completely independent of y and so places no restriction on that variable for constant energy. The condition on U means that it has a *confining* property. To ensure bounded solutions in systems that do not have a confining potential, we should add other assumptions, for example the assumption that the position vectors of each atom are restricted by periodic boundary conditions to lie on a three-dimensional torus (see Section 1.6).

1.5.1 The Flow Map

Consider now the initial value problem

$$\dot{\boldsymbol{z}} = \boldsymbol{f}(\boldsymbol{z}), \qquad \boldsymbol{z}(0) = \boldsymbol{\zeta}, \tag{1.5}$$

in a m-dimensional space. If we assume that $\boldsymbol{f}$ corresponds to a molecular Hamiltonian system satisfying the assumptions of the existence and uniqueness result of the previous subsection, then we may define a mapping from a point in phase space to the point t units later along the time-evolution starting from the initial point. We refer to this map as the *flow map* and denote it by $\mathscr{F}_t$. $\mathscr{F}_t(\boldsymbol{\zeta}) = \boldsymbol{z}(t)$ solves the initial value problem (1.5). It may equally describe the evolution of a point with time, or the evolution of a set of points with time.

The classical systems treated here can be solved forward or backward in time. Observe that $\mathscr{F}_{-t}\mathscr{F}_t = \mathrm{Id}$ (the identity map), thus the flow map is invertible and, indeed, the family of flow maps defined for different values of t form an Abelian group under the operation of composition ($\mathscr{F}_t\mathscr{F}_s = \mathscr{F}_s\mathscr{F}_t = \mathscr{F}_{t+s}$). The flow map of a Hamiltonian system conserves its Hamiltonian, thus

$$H(\mathscr{F}_t(\boldsymbol{\zeta})) \equiv H(\boldsymbol{\zeta}).$$

(This equation just expresses the conservation of energy.)

The flow map provides an easy notation for describing solutions (even in cases where it is difficult or impossible to compute it exactly). When we speak of numerical methods for solving differential equations, we generally mean methods that approximate the flow map in a suitable sense.

Despite the existence of a well-defined flow map, we cannot solve most systems of differential equations analytically (even Hamiltonian ones) so we cannot usually write out a formula for the flow map. An exception is the harmonic oscillator,

$$\dot{q} = p, \qquad \dot{p} = -\Omega^2 q,$$